Section A

1 **(a)** **(i)** Newton's first law of motion states that a body continues at rest or at constant / uniform velocity unless acted on by a <u>resultant</u> (external) force. [A1]

(ii) resultant force (in any direction) is zero [B1]

resultant moment / torque (about any axis) is zero [B1]

(b) **(i)** Using principle of moment and taking moment about the bottom of ladder: [B1]

clockwise moment = anticlockwise moment

$$\mathrm{N} \times \mathrm{L} \sin 60^\circ = \mathrm{W} \times \frac{L}{2} \cos 60^\circ$$

$$\mathrm{N} \times \mathrm{L} \sin 60^\circ = 80 \times \frac{L}{2} \cos 60^\circ \text{ [B1]}$$

N = 23 N [A0]

(ii) Resolve vertically: $\uparrow$ Y = W = 80 N

Resolve horizontally $\leftarrow$ X = N = 23 N [C1 for both equations]

force R = $\sqrt{X^2 + Y^2} = \sqrt{23^2 + 80^2}$ = 83 N [A1]

angle R makes with floor = $\tan^{-1}\left(\frac{Y}{X}\right) = \tan^{-1}\left(\frac{80}{23}\right) = 74^\circ$

Direction: 74° clockwise above horizontal [A1]

(iii) **1.** (Due to the person's weight) there is now <u>greater downward force on the ladder</u>, and so (to maintain equilibrium) the floor exerts a <u>larger</u> upward vertical force on the ladder. [A1]

2. (Due to the person's weight) there is now a greater anticlockwise moment about